

Extensive Solved Problem Set

Calculus and Ordinary Differential Equations (AS1109)

Section H – Problems from the single-session tutorial and extra-problem bank

Who this booklet is for

This booklet is designed especially so that a student who has joined the course late and remembers mainly **Class XII mathematics** can still follow the reasoning. We therefore do not jump directly to formulas. Before each solution, we identify the idea, recall any prerequisite, and explain what each derivative or limit is actually doing.

Problems included

The booklet contains the **last three problems from the Section H single-session tutorial** (Q5–Q7) followed by **all seven optional extra problems** (E1–E7). The original problem statements are preserved; only the explanations and solution detail have been expanded.

How to study from this booklet

For each problem, try the question for two or three minutes before reading the solution. Then follow the same five-stage pattern:

What is being asked? $\longrightarrow$ What do I already know? $\longrightarrow$ Work step by step $\longrightarrow$ Interpret $\longrightarrow$ Check

A short bridge from Class XII to multivariable calculus

1. From ordinary derivatives to partial derivatives

In Class XII, a function such as $y = f(x)$ has one independent variable. Its derivative $f'(x)$ measures how the output changes when x changes.

Now suppose $z = f(x, y)$. There are two independent inputs. A **partial derivative** asks us to vary one input while temporarily holding the other fixed:

$$f_x = \frac{\partial f}{\partial x} : \text{change in } f \text{ when } x \text{ changes and } y \text{ is fixed,}$$

$$f_y = \frac{\partial f}{\partial y} : \text{change in } f \text{ when } y \text{ changes and } x \text{ is fixed.}$$

So partial differentiation is ordinary differentiation with a clear instruction about which variable is allowed to move.

2. Second partial derivatives and the Hessian

We can differentiate again. From f_x and f_y we obtain

$$f_{xx}, \quad f_{xy}, \quad f_{yx}, \quad f_{yy}.$$

The first subscript tells us which derivative was taken first in the notation used in these notes: for example, $f_{xy} = \partial(f_x)/\partial y$.

The four second partials are arranged in the **Hessian matrix**

$$H_f = \begin{pmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{pmatrix}.$$

For sufficiently smooth functions, the mixed partials are equal: $f_{xy} = f_{yx}$. This is a theorem, not an automatic algebra rule.

3. Multivariable limits: why paths matter

For a one-variable limit $x \rightarrow a$, we approach from left or right. For $(x, y) \rightarrow (0, 0)$, there are infinitely many ways to approach the origin: along $y = 0$, $x = 0$, $y = x$, $y = x^2$, curves, spirals, and so on. Therefore, if two different paths give two different limiting values, the two-variable limit **cannot exist**. This is one of the most useful tests in this chapter.

4. Implicit differentiation: the chain-rule reminder

If an equation such as

$$F(x, y, z) = 0$$

defines z as a function of x and y , then z is not a constant. When differentiating with respect to x ,

$$\frac{\partial}{\partial x}(z^3) = 3z^2 z_x.$$

This is just the familiar Class XII chain rule, with z_x replacing dz/dx .

5. Laplace and wave equations

In this course, you are not yet solving partial differential equations. You are mainly asked to **recognise or verify** them by computing derivatives and substituting them into the stated equation. Two important forms are

$$\boxed{u_{xx} + u_{yy} = 0} \quad (\text{Laplace equation}),$$

$$\boxed{v_{tt} = c^2 v_{xx}} \quad (\text{one-dimensional wave equation}).$$

Problem 1: First, higher and mixed partial derivatives; Hessian

Original tutorial Q5

For

$$g(x, y) = e^{xy} + \sin(x + y) + x^2y,$$

find all first and second partial derivatives, write the Hessian matrix, discuss whether it is symmetric, and interpret g_x and g_y geometrically.

Before we solve

We will differentiate one term at a time. Remember:

$$\frac{\partial}{\partial x} e^{xy} = ye^{xy}, \quad \frac{\partial}{\partial y} e^{xy} = xe^{xy},$$

because when differentiating with respect to x , y is temporarily treated as a constant, and vice versa.

Step-by-step solution

Step 1: Find g_x .

$$g = e^{xy} + \sin(x + y) + x^2y.$$

Differentiate with respect to x , holding y fixed:

$$\frac{\partial}{\partial x}(e^{xy}) = ye^{xy},$$

$$\frac{\partial}{\partial x} \sin(x + y) = \cos(x + y) \cdot \frac{\partial}{\partial x}(x + y) = \cos(x + y),$$

$$\frac{\partial}{\partial x}(x^2y) = 2xy.$$

Therefore

$$g_x = ye^{xy} + \cos(x + y) + 2xy.$$

Step 2: Find g_y . Differentiate the same original function with respect to y , now holding x fixed:

$$\frac{\partial}{\partial y}(e^{xy}) = xe^{xy},$$

$$\frac{\partial}{\partial y} \sin(x + y) = \cos(x + y),$$

$$\frac{\partial}{\partial y}(x^2y) = x^2.$$

Hence

$$g_y = xe^{xy} + \cos(x + y) + x^2.$$

Step 3: Differentiate again to obtain the pure second partials.

$$\begin{aligned} g_{xx} &= \frac{\partial}{\partial x}(ye^{xy} + \cos(x + y) + 2xy) \\ &= y^2e^{xy} - \sin(x + y) + 2y, \end{aligned}$$

so

$$g_{xx} = y^2e^{xy} - \sin(x + y) + 2y.$$

Similarly,

$$\begin{aligned} g_{yy} &= \frac{\partial}{\partial y} (xe^{xy} + \cos(x+y) + x^2) \\ &= x^2 e^{xy} - \sin(x+y), \end{aligned}$$

so

$$g_{yy} = x^2 e^{xy} - \sin(x+y).$$

Step 4: Find the mixed partials. Starting with g_x and differentiating with respect to y ,

$$\begin{aligned} g_{xy} &= \frac{\partial}{\partial y} (ye^{xy} + \cos(x+y) + 2xy) \\ &= \underbrace{e^{xy} + xye^{xy}}_{\text{product rule on } ye^{xy}} - \sin(x+y) + 2x. \end{aligned}$$

Thus

$$g_{xy} = e^{xy}(1+xy) - \sin(x+y) + 2x.$$

Starting with g_y and differentiating with respect to x gives

$$g_{yx} = e^{xy}(1+xy) - \sin(x+y) + 2x.$$

Step 5: Put the second partials into the Hessian.

$$H_g(x, y) = \begin{pmatrix} y^2 e^{xy} - \sin(x+y) + 2y & e^{xy}(1+xy) - \sin(x+y) + 2x \\ e^{xy}(1+xy) - \sin(x+y) + 2x & x^2 e^{xy} - \sin(x+y) \end{pmatrix}.$$

Because the two off-diagonal entries are equal, the Hessian is symmetric.

Why this works

All terms in g are smooth functions: exponentials, sine, and polynomials. Their relevant partial derivatives are continuous everywhere. Therefore the hypotheses of the Mixed Derivative Theorem are satisfied, so $g_{xy} = g_{yx}$ is expected.

Takeaway

For the surface $z = g(x, y)$, $g_x(a, b)$ is the slope of the trace obtained by cutting the surface with the plane $y = b$. Similarly, $g_y(a, b)$ is the slope of the trace in the plane $x = a$. In plain language: g_x measures the local rate of change in the x -direction, and g_y in the y -direction.

Common pitfall

When differentiating ye^{xy} with respect to y , use the product rule. A common mistake is to write only xe^{xy} and forget the derivative of the leading factor y .

Problem 2: Implicit differentiation with two independent variables

Original tutorial Q6

If

$$xy + z^3y - 2xz = 0$$

defines $z = z(x, y)$, derive z_x and z_y and evaluate them at $(1, 1, 1)$.

Before we solve

Although the equation contains x, y, z , only x and y are independent. The variable z changes when x or y changes. Therefore every derivative of a term containing z must include a chain-rule factor z_x or z_y .

Step-by-step solution

Step 1: Check that the given point is on the surface. At $(1, 1, 1)$,

$$xy + z^3y - 2xz = 1 + 1 - 2 = 0.$$

So the point is valid.

Step 2: Differentiate with respect to x . While differentiating with respect to x , treat y as a constant:

$$\frac{\partial}{\partial x}(xy) = y.$$

For z^3y , the factor y is constant but z depends on x :

$$\frac{\partial}{\partial x}(z^3y) = y(3z^2z_x).$$

For $-2xz$, use the product rule because both x and z vary with x :

$$\frac{\partial}{\partial x}(-2xz) = -2z - 2xz_x.$$

Thus

$$y + 3z^2yz_x - 2z - 2xz_x = 0.$$

Collect the z_x terms:

$$(3z^2y - 2x)z_x = 2z - y.$$

Hence

$$z_x = \frac{2z - y}{3z^2y - 2x}.$$

At $(1, 1, 1)$,

$$z_x = \frac{2(1) - 1}{3(1)^2(1) - 2(1)} = \frac{1}{1} = \boxed{1}.$$

Step 3: Differentiate with respect to y . Now treat x as a constant:

$$\frac{\partial}{\partial y}(xy) = x.$$

For z^3y , both z and y vary with y , so use the product rule:

$$\frac{\partial}{\partial y}(z^3y) = 3z^2z_y y + z^3.$$

For $-2xz$, x is constant:

$$\frac{\partial}{\partial y}(-2xz) = -2xz_y.$$

Therefore

$$x + 3z^2yz_y + z^3 - 2xz_y = 0.$$

Collect the z_y terms:

$$(3z^2y - 2x)z_y = -(x + z^3).$$

So

$$z_y = -\frac{x + z^3}{3z^2y - 2x}.$$

At $(1, 1, 1)$,

$$z_y = -\frac{1 + 1}{3 - 2} = \boxed{-2}.$$

Why this works

This is exactly the Class XII chain rule in multivariable notation. If $z = z(x, y)$, then z^3 is a composite function. Differentiating z^3 with respect to x gives $3z^2z_x$; with respect to y it gives $3z^2z_y$.

Final answer

$$z_x = \frac{2z - y}{3z^2y - 2x}, \quad z_y = -\frac{x + z^3}{3z^2y - 2x}$$

and at $(1, 1, 1)$,

$$\boxed{z_x = 1, \quad z_y = -2.}$$

Common pitfall

Do not differentiate z^3 as if z were constant. The entire point of implicit differentiation is that z is changing even though no explicit formula $z = f(x, y)$ has been written down.

Problem 3: Verifying Laplace's equation and the wave equation

Original tutorial Q7

(a) Verify that

$$u(x, y) = e^{-x} \cos y$$

satisfies

$$u_{xx} + u_{yy} = 0.$$

(b) For

$$v(x, t) = \cos(kx - \omega t),$$

determine the relation between k, ω, c needed for

$$v_{tt} = c^2 v_{xx}.$$

Before we solve

A verification problem has a simple recipe: **differentiate, substitute, simplify**. You do not need to solve the PDE from scratch.

Step-by-step solution

Part (a): Laplace equation.

First differentiate with respect to x :

$$u_x = -e^{-x} \cos y.$$

Differentiate once more with respect to x :

$$u_{xx} = e^{-x} \cos y.$$

Now differentiate with respect to y :

$$u_y = -e^{-x} \sin y,$$

so

$$u_{yy} = -e^{-x} \cos y.$$

Add the two second derivatives:

$$u_{xx} + u_{yy} = e^{-x} \cos y - e^{-x} \cos y = 0.$$

Therefore

$$\boxed{u_{xx} + u_{yy} = 0},$$

so u satisfies Laplace's equation.

Part (b): Wave equation.

Let

$$\phi = kx - \omega t,$$

so $v = \cos \phi$.

Differentiate twice with respect to x :

$$v_x = -k \sin \phi,$$

$$v_{xx} = -k^2 \cos \phi = -k^2 v.$$

Differentiate twice with respect to t :

$$v_t = \omega \sin \phi,$$

$$v_{tt} = -\omega^2 \cos \phi = -\omega^2 v.$$

Substitute these into the wave equation:

$$-\omega^2 v = c^2(-k^2 v).$$

For a nonzero wave, cancel $-v$:

$$\omega^2 = c^2 k^2.$$

Therefore

$$\boxed{\omega^2 = c^2 k^2}.$$

For positive k, ω, c this is usually written

$$\boxed{\omega = ck}.$$

Takeaway

Laplace's equation appears in steady-state or equilibrium models such as temperature in a homogeneous plate and electrostatic potential in a charge-free region. The wave equation relates curvature in space (v_{xx}) to acceleration in time (v_{tt}); the constant c represents wave speed.

Problem 4: Domain, boundary, openness and range

Extra problem E1

Consider

$$f(x, y) = \frac{1}{\sqrt{x^2 + y^2 - 9}}.$$

- Determine and sketch the domain.
- Find its boundary and state whether the domain is open or closed.
- Find the range of f .

Before we solve

Three familiar Class XII restrictions are useful:

$$\sqrt{A} \text{ requires } A \geq 0, \quad \frac{1}{A} \text{ requires } A \neq 0, \quad \frac{1}{\sqrt{A}} \text{ therefore requires } A > 0.$$

Step-by-step solution

The expression under the square root is

$$x^2 + y^2 - 9.$$

Because this square root is in the denominator, it must be strictly positive:

$$x^2 + y^2 - 9 > 0.$$

Thus

$$x^2 + y^2 > 9.$$

The equation $x^2 + y^2 = 9$ is a circle of radius 3. The inequality $x^2 + y^2 > 9$ means all points **outside** that circle, with the circle itself excluded.

Therefore

$$D = \{(x, y) : x^2 + y^2 > 9\}.$$

Its boundary is

$$x^2 + y^2 = 9.$$

The domain is **open**: every point in it has some small surrounding disk that remains outside the circle. It is not closed because it does not contain its boundary.

For the range, write $r^2 = x^2 + y^2$. Then $r > 3$ and

$$f = \frac{1}{\sqrt{r^2 - 9}} > 0.$$

As $r \rightarrow 3^+$, the denominator tends to 0^+ , so $f \rightarrow \infty$. As $r \rightarrow \infty$, $f \rightarrow 0^+$, but never actually reaches 0. Thus

$$\text{Range}(f) = (0, \infty).$$

Why this works

The range is easier if you think of moving radially outward from the circle. Immediately outside the boundary the denominator is tiny, so the function is huge. Very far away the denominator is huge, so the function is small and positive.

Problem 5: Contours and steepness

Extra problem E2

A temperature field is

$$T(x, y) = 100 - x^2 - 4y^2.$$

- Find the level curves $T = 96$ and $T = 84$ and identify their shapes.
- Near the origin, does temperature change more rapidly in the x -direction or the y -direction? Explain from the contour geometry.

Before we solve

A **level curve** means “set the output equal to a constant”. For example, $T = 96$ means all points (x, y) where the temperature is 96.

Step-by-step solution

Step 1: Find the level curve $T = 96$.

$$100 - x^2 - 4y^2 = 96.$$

Move terms:

$$x^2 + 4y^2 = 4.$$

Divide by 4:

$$\frac{x^2}{4} + y^2 = 1.$$

This is an ellipse with semiaxis 2 along x and semiaxis 1 along y .

Step 2: Find the level curve $T = 84$.

$$100 - x^2 - 4y^2 = 84$$

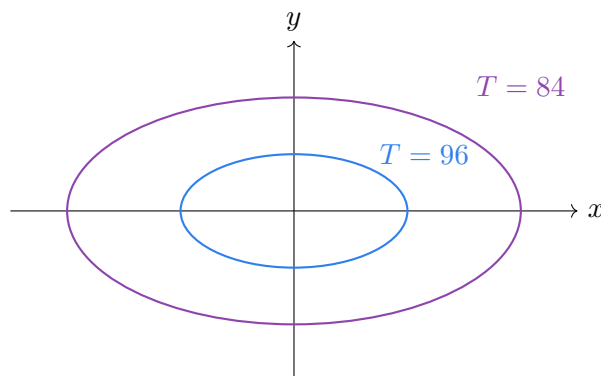
gives

$$x^2 + 4y^2 = 16,$$

or

$$\frac{x^2}{16} + \frac{y^2}{4} = 1.$$

This is an ellipse with semiaxis 4 along x and semiaxis 2 along y .



Interpreting the spacing

To reach the same temperature drop from 100 to 96:

- along the x -axis, set $y = 0$: $x^2 = 4$, so $|x| = 2$;
- along the y -axis, set $x = 0$: $4y^2 = 4$, so $|y| = 1$.

So the same change in temperature occurs over only half as much distance in the y -direction. The contours are therefore more tightly packed vertically, indicating more rapid change in the y -direction.

Common pitfall

At the exact origin, $T_x(0, 0) = T_y(0, 0) = 0$, so the first-order rates are both zero there. The statement “more rapid in the y -direction” refers to the nearby contour geometry / finite change away from the maximum, where the y coefficient 4 makes the field fall off faster.

Problem 6: A two-variable limit where the right path matters

Extra problem E3

Determine whether

$$\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 y}{x^4 + y^2}$$

exists. Try $y = 0$ and then a path that makes y comparable with x^2 .

Before we solve

The denominator contains x^4 and y^2 . A clever path makes these two terms have the same order. Since y^2 will equal x^4 when $y = x^2$, that path is a natural choice.

Step-by-step solution

Path 1: $y = 0$.

$$\frac{x^2(0)}{x^4 + 0} = 0.$$

Therefore along the x -axis,

$$\lim_{x \rightarrow 0} \frac{x^2(0)}{x^4} = 0.$$

Path 2: $y = x^2$. Substitute $y = x^2$:

$$\frac{x^2(x^2)}{x^4 + (x^2)^2} = \frac{x^4}{x^4 + x^4} = \frac{1}{2}$$

for $x \neq 0$. Hence along this path,

$$\lim_{x \rightarrow 0} \frac{x^2(x^2)}{x^4 + x^4} = \frac{1}{2}.$$

The two path limits are different:

$$0 \neq \frac{1}{2}.$$

Therefore the full two-variable limit does not exist.

Final answer

$$\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 y}{x^4 + y^2} \text{ does not exist.}$$

Takeaway

When a denominator contains competing powers such as $x^4 + y^2$, try a path that balances them: here $y = x^2$ because then $y^2 = x^4$.

Problem 7: Physical meaning of partial derivatives

Extra problem E4

The temperature on a thin plate is

$$T(x, y) = 100e^{-0.1(x^2+y^2)}.$$

Find T_x and T_y , evaluate them at $(2, 1)$, and interpret the signs physically.

Before we solve

This is an application of the chain rule you already know. If

$$T = 100e^u, \quad u = -0.1(x^2 + y^2),$$

then $T_x = 100e^u u_x$ and $T_y = 100e^u u_y$.

Step-by-step solution

For the x derivative,

$$u_x = -0.1(2x) = -0.2x.$$

Therefore

$$T_x = 100e^{-0.1(x^2+y^2)}(-0.2x) = \boxed{-20xe^{-0.1(x^2+y^2)}}.$$

Similarly,

$$u_y = -0.2y,$$

so

$$T_y = \boxed{-20ye^{-0.1(x^2+y^2)}}.$$

At $(2, 1)$,

$$x^2 + y^2 = 4 + 1 = 5,$$

so the exponent is $-0.1(5) = -0.5$. Hence

$$\boxed{T_x(2, 1) = -40e^{-1/2}},$$

$$\boxed{T_y(2, 1) = -20e^{-1/2}}.$$

Numerically, since $e^{-1/2} \approx 0.6065$,

$$T_x(2, 1) \approx -24.26, \quad T_y(2, 1) \approx -12.13.$$

Takeaway

The negative signs mean that, from the point $(2, 1)$, moving a small distance in the positive x direction or positive y direction decreases the temperature. The magnitude of T_x is twice the magnitude of T_y , so the temperature is changing twice as rapidly per unit distance in the $+x$ direction at this point.

Problem 8: Mixed partials and the Hessian – a second example

Extra problem E5

Let

$$f(x, y) = x^2 e^y + y^3.$$

Find $f_{xx}, f_{xy}, f_{yx}, f_{yy}$ and write $H_f(x, y)$. State why $f_{xy} = f_{yx}$ is justified here.

Before we solve

Before finding second derivatives, first find f_x and f_y . This keeps the work organised and reduces errors.

Step-by-step solution

First partial derivatives:

$$f_x = 2xe^y,$$

$$f_y = x^2 e^y + 3y^2.$$

Now differentiate again:

$$f_{xx} = \frac{\partial}{\partial x}(2xe^y) = 2e^y,$$

$$f_{xy} = \frac{\partial}{\partial y}(2xe^y) = 2xe^y,$$

$$f_{yx} = \frac{\partial}{\partial x}(x^2 e^y + 3y^2) = 2xe^y,$$

$$f_{yy} = \frac{\partial}{\partial y}(x^2 e^y + 3y^2) = x^2 e^y + 6y.$$

Therefore

$$H_f(x, y) = \begin{pmatrix} 2e^y & 2xe^y \\ 2xe^y & x^2 e^y + 6y \end{pmatrix}.$$

Why this works

The functions appearing here are polynomials and exponentials, so the relevant partial derivatives are continuous everywhere. The Mixed Derivative Theorem therefore guarantees $f_{xy} = f_{yx}$.

Problem 9: Implicit differentiation on a sphere

Extra problem E6

The sphere

$$x^2 + y^2 + z^2 = 9$$

defines z locally as a function of x, y near $(1, 2, 2)$. Find z_x and z_y at this point.

Before we solve

The equation says that (x, y, z) must remain on a sphere of radius 3. If x changes while y is fixed, z must adjust to keep the equation true.

Step-by-step solution

Find z_x . Differentiate with respect to x , holding y fixed:

$$2x + 0 + 2zz_x = 0.$$

So

$$2zz_x = -2x,$$

$$z_x = -\frac{x}{z}.$$

At $(1, 2, 2)$,

$$z_x = -\frac{1}{2}.$$

Find z_y . Differentiate with respect to y , holding x fixed:

$$0 + 2y + 2zz_y = 0.$$

Hence

$$z_y = -\frac{y}{z}.$$

At $(1, 2, 2)$,

$$z_y = -\frac{2}{2} = -1.$$

Why this works

The negative signs match the geometry. On the upper part of the sphere, if you increase x or y near this point, z must decrease to keep $x^2 + y^2 + z^2$ equal to 9.

Problem 10: First partial derivatives can exist even when continuity fails

Extra problem E7

Define

$$f(x, y) = \begin{cases} \frac{xy}{x^2 + y^2}, & (x, y) \neq (0, 0), \\ 0, & (x, y) = (0, 0). \end{cases}$$

Show directly from the definition that $f_x(0, 0)$ and $f_y(0, 0)$ exist. Then show that f is not continuous at $(0, 0)$.

Before we solve

This problem separates two ideas:

- a partial derivative checks change along one coordinate direction;
- continuity checks what happens from *all* directions.

So it is possible for the coordinate-direction tests to behave nicely while another path behaves badly.

Step-by-step solution

Step 1: Compute $f_x(0, 0)$ from the definition.

$$f_x(0, 0) = \lim_{h \rightarrow 0} \frac{f(h, 0) - f(0, 0)}{h}.$$

For $h \neq 0$,

$$f(h, 0) = \frac{h \cdot 0}{h^2 + 0} = 0.$$

Also $f(0, 0) = 0$. Therefore

$$f_x(0, 0) = \lim_{h \rightarrow 0} \frac{0 - 0}{h} = 0.$$

Hence

$$\boxed{f_x(0, 0) = 0}.$$

Step 2: Compute $f_y(0, 0)$.

$$f_y(0, 0) = \lim_{k \rightarrow 0} \frac{f(0, k) - f(0, 0)}{k}.$$

For $k \neq 0$,

$$f(0, k) = \frac{0 \cdot k}{0 + k^2} = 0.$$

Thus

$$f_y(0, 0) = \lim_{k \rightarrow 0} \frac{0}{k} = 0,$$

so

$$\boxed{f_y(0, 0) = 0}.$$

Step 3: Test continuity using the path $y = x$. For $x \neq 0$,

$$f(x, x) = \frac{x \cdot x}{x^2 + x^2} = \frac{x^2}{2x^2} = \frac{1}{2}.$$

Therefore

$$\lim_{x \rightarrow 0} f(x, x) = \frac{1}{2}.$$

But

$$f(0,0) = 0.$$

Since the function approaches $1/2$ along $y = x$ instead of approaching its value 0,

$$f \text{ is not continuous at } (0,0).$$

Why this works

The partial derivatives at the origin examined only the two coordinate axes. Along both axes the numerator xy is zero, so everything looks harmless. Continuity is stricter: the diagonal path $y = x$ reveals a completely different behaviour.

Final answer

$$f_x(0,0) = 0, \quad f_y(0,0) = 0,$$

but

$$f \text{ is not continuous at } (0,0).$$

Therefore existence of both first partial derivatives does **not** imply continuity, and hence certainly does not by itself imply differentiability.

One-look revision map

Problem type	First question to ask yourself
Domain/range	What restrictions come from square roots, denominators, or logarithms?
Level curves/contours	What equation do I get after setting the output equal to a constant?
Two-variable limit	Can I find two paths that give different values?
Partial derivative	Which variable is changing, and which one is being held fixed?
Second/mixed partials	Which first partial should I differentiate next?
Hessian	Have I arranged $f_{xx}, f_{xy}, f_{yx}, f_{yy}$ correctly?
Implicit derivative	Did I attach z_x or z_y whenever I differentiated a z term?
Laplace/wave verification	Did I compute the required second derivatives and substitute them into the equation?
Continuity counterexample	Do the coordinate axes hide behaviour that another path reveals?

A final study habit

When a multivariable question looks unfamiliar, do not start by searching for a formula. First translate the question into an action: *restrict the domain, hold one variable fixed, try a path, differentiate and substitute, or use the chain rule*. Once the action is identified, most of the algebra is built from Class XI–XII skills.